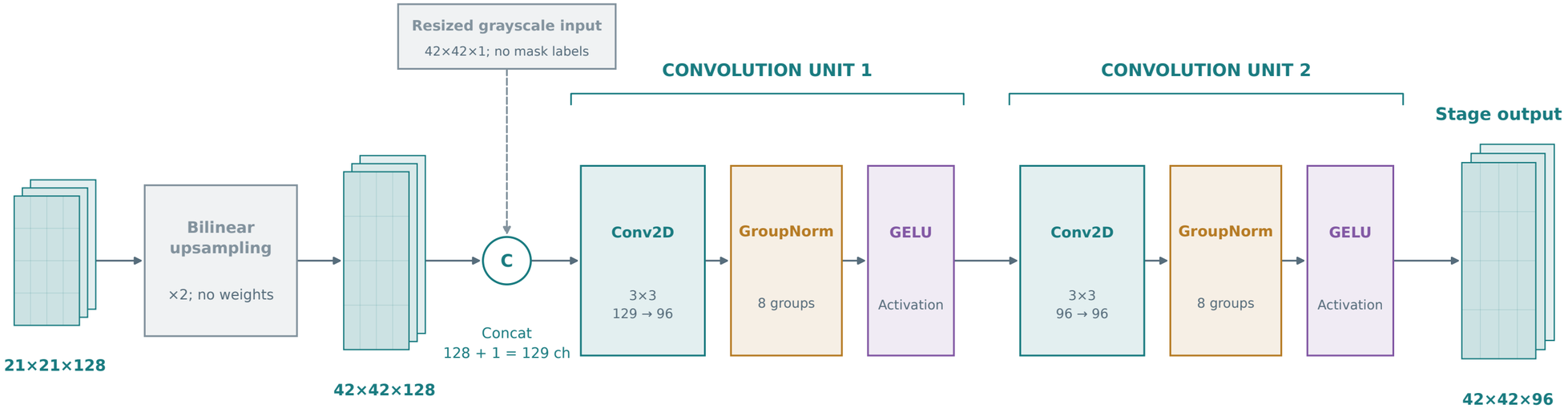
What is inside an added decoder stage?

Stage 1 shown layer by layer. The same operation sequence is used in all four stages.



What learns?

Conv: local spatial filters
GroupNorm: scale and offset
GELU: fixed activation
Upsampling: fixed interpolation

All four stages: spatial resolution increases while channel width decreases

Stage	Input feature map	After ×2 resize + gray concat	Conv 1	Conv 2	Output feature map
1	21×21×128	42×42×129	3×3: 129 → 96	3×3: 96 → 96	42×42×96
2	42×42×96	84×84×97	3×3: 97 → 64	3×3: 64 → 64	84×84×64
3	84×84×64	168×168×65	3×3: 65 → 32	3×3: 32 → 32	168×168×32
4	168×168×32	336×336×33	3×3: 33 → 24	3×3: 24 → 24	336×336×24

Final pixel prediction:

1×1 Conv: 24 → 3
One logit for each pixel class

→

Softmax over classes
No learned parameters

→

336×336 probability map per class
Normal / crack / other defect

Conv stride = 1; padding = 1. Every 3×3 convolution is followed by GroupNorm(8) and GELU. Feature volumes are schematic.